复变函数第八次作业

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5.15. 计算下列积分:

(2) $\displaystyle\int_0^{+\infty}\frac{\mathrm{d}x}{1+x^p}\,(p\in\mathbb{R}_{>1})$;

(4) $\displaystyle\int_0^{2\pi}\frac{\mathrm{d}\theta}{1-2a\cos\theta+a^2}\,(a\in\mathbb{R},|a|<1)$;

(6) $\displaystyle\int_0^{+\infty}\frac{\cos x-\mathrm{e}^{-x}}{x}\mathrm{d}x$;

(8) $\displaystyle\int_0^{+\infty}x^{p-1}\cos x\mathrm{d}x\,(p\in\mathbb{R},0<p<1)$;

(10) $\displaystyle\int_0^{+\infty}\frac{x\sin x}{(x^2+1)^2}\mathrm{d}x$.

解答:

(2) 做换元 $x=t^{\frac{1}{p}},t\geqslant 0$ 再结合**例5**得

$$\int_0^{+\infty}\frac{\mathrm{d}x}{1+x^p}=\frac{1}{p}\int_0^{+\infty}\frac{t^{\frac{1}{p}-1}}{1+t}\mathrm{d}t=\frac{\pi}{p\sin\frac{\pi}{p}}.$$

(4) 做换元 $z=\mathrm{e}^{\mathrm{i}\theta}$ 得

$$\begin{aligned}\int_0^{2\pi}\frac{\mathrm{d}\theta}{1-2a\cos\theta+a^2}&=\int_{|z|=1}\frac{1}{1-a\left(z+\frac{1}{z}\right)+a^2}\cdot\frac{\mathrm{d}z}{\mathrm{i}z}\\&=\mathrm{i}\int_{|z|=1}\frac{\mathrm{d}z}{az^2-(a^2+1)z+a}\\&=\mathrm{i}\int_{|z|=1}\frac{\mathrm{d}z}{(az-1)(z-a)}\\&=\mathrm{i}\cdot 2\pi\mathrm{i}\mathrm{Res}\left(\frac{1}{(az-1)(z-a)},a\right)\\&=\frac{2\pi}{1-a^2}.\end{aligned}$$

(6) 取 $f(z)=\dfrac{\mathrm{e}^{\mathrm{i}z}-\mathrm{e}^{-z}}{z}$, 以及四分之一扇形回路 $[r,R]\to\gamma_R\to[\mathrm{i}R,\mathrm{i}r]\to\gamma_r$.

由 **Cauchy 定理**得

$$\left(\int_{[r,R]}+\int_{\gamma_R}+\int_{[\mathrm{i}R,\mathrm{i}r]}+\int_{\gamma_r}\right)f(z)\mathrm{d}z=0.$$

当 $r\to 0^+,R\to+\infty$ 时,

$$\begin{aligned}\left|\int_{\gamma_R}f(z)\mathrm{d}z\right|&\leqslant\int_0^{\frac{\pi}{2}}\frac{\mathrm{e}^{-R\sin\theta}+\mathrm{e}^{-R\cos\theta}}{R}\cdot R\mathrm{d}\theta\\&=2\int_0^{\frac{\pi}{2}}\mathrm{e}^{-R\sin\theta}\mathrm{d}\theta\leqslant 2\int_0^{\frac{\pi}{2}}\mathrm{e}^{-\frac{2R}{\pi}\theta}\mathrm{d}\theta\\&=\frac{\pi}{R}\left(1-\mathrm{e}^{-R}\right)\to 0,\end{aligned}$$

$$\int_{\gamma_r} f(z)\mathrm{d}z = \int_{\gamma_r} \frac{(1+\mathrm{i}z+o(z))-(1-z+o(z))}{z}\mathrm{d}z$$
$$= \int_{\gamma_r} (\mathrm{i}+1+o(1))\mathrm{d}z = \frac{\pi r}{2}(\mathrm{i}+1+o(1)) \to 0,$$

$$\int_{[r,R]} f(z)\mathrm{d}z \to \int_0^{+\infty} \frac{\mathrm{e}^{\mathrm{i}x}-\mathrm{e}^{-x}}{x}\mathrm{d}x,$$

$$\int_{[\mathrm{i}R,\mathrm{i}r]} f(z)\mathrm{d}z \to \int_{+\infty}^0 \frac{\mathrm{e}^{\mathrm{i}\cdot\mathrm{i}x}-\mathrm{e}^{-\mathrm{i}x}}{\mathrm{i}x}\mathrm{d}(\mathrm{i}x) = \int_0^{+\infty} \frac{\mathrm{e}^{-\mathrm{i}x}-\mathrm{e}^{-x}}{x}\mathrm{d}x,$$

故

$$\int_0^{+\infty} \frac{\mathrm{e}^{\mathrm{i}x}-\mathrm{e}^{-x}}{x}\mathrm{d}x + \int_0^{+\infty} \frac{\mathrm{e}^{-\mathrm{i}x}-\mathrm{e}^{-x}}{x}\mathrm{d}x = 0,$$

即

$$\int_0^{+\infty} \frac{\cos x-\mathrm{e}^{-x}}{x}\mathrm{d}x = 0.$$

(8) 取 $f(z)=z^{p-1}\mathrm{e}^{\mathrm{i}z}$ 在实轴上为实值的分支, 以及四分之一扇形回路 $[r,R]\to\gamma_R\to[\mathrm{i}R,\mathrm{i}r]\to\gamma_r$.

由 **Cauchy 定理**得

$$\left(\int_{[r,R]}+\int_{\gamma_R}+\int_{[\mathrm{i}R,\mathrm{i}r]}+\int_{\gamma_r}\right) f(z)\mathrm{d}z = 0.$$

当 $r\to 0^+, R\to+\infty$ 时,

$$\left|\int_{\gamma_R} f(z)\mathrm{d}z\right| \leqslant \int_0^{\frac{\pi}{2}} R^{p-1}\mathrm{e}^{-R\sin\theta}\cdot R\mathrm{d}\theta$$
$$\leqslant R^p\int_0^{\frac{\pi}{2}} \mathrm{e}^{-\frac{2R}{\pi}\theta}\mathrm{d}\theta = \frac{\pi}{2R^{1-p}}\left(1-\mathrm{e}^{-R}\right) \to 0,$$

$$\left|\int_{\gamma_r} f(z)\mathrm{d}z\right| \leqslant r^{p-1}\cdot\frac{\pi r}{2} = \frac{\pi}{2}r^p \to 0,$$

$$\int_{[r,R]} f(z)\mathrm{d}z \to \int_0^{+\infty} x^{p-1}\mathrm{e}^{\mathrm{i}x}\mathrm{d}x,$$

$$\int_{[\mathrm{i}R,\mathrm{i}r]} f(z)\mathrm{d}z \to \int_{+\infty}^0 (\mathrm{i}x)^{p-1}\mathrm{e}^{\mathrm{i}\cdot\mathrm{i}x}\mathrm{d}(\mathrm{i}x) = -\mathrm{i}^p\int_0^{+\infty} x^{p-1}\mathrm{e}^{-x}\mathrm{d}x = -\mathrm{i}^p\Gamma(p),$$

故

$$\int_0^{+\infty} x^{p-1}\cos x\mathrm{d}x = \mathrm{Re}\left(\int_0^{+\infty} x^{p-1}\mathrm{e}^{\mathrm{i}x}\mathrm{d}x\right) = \mathrm{Re}\left(\mathrm{i}^p\Gamma(p)\right) = \cos\left(\frac{p\pi}{2}\right)\Gamma(p).$$

(10) 取 $f(z)=\dfrac{z\mathrm{e}^{\mathrm{i}z}}{(z^2+1)^2}$, 以及半圆回路 $[r,R]\to\gamma_R\to[-R,-r]\to\gamma_r$.

由 **Cauchy 定理**得

$$\left(\int_{[r,R]}+\int_{\gamma_R}+\int_{[\mathrm{i}R,\mathrm{i}r]}+\int_{\gamma_r}\right) f(z)\mathrm{d}z = 2\pi\mathrm{i}\mathrm{Res}(f,\mathrm{i}).$$

当 $r\to 0^+, R\to+\infty$ 时,

$$\left|\int_{\gamma_R} f(z)\mathrm{d}z\right| \leqslant \int_0^{\pi} \frac{R}{(R^2-1)^2}\mathrm{e}^{-R\sin\theta}\cdot R\mathrm{d}\theta$$
$$= \frac{2R^2}{(R^2-1)^2}\int_0^{\frac{\pi}{2}} \mathrm{e}^{-R\sin\theta}\mathrm{d}\theta$$
$$\leqslant \frac{2R^2}{(R^2-1)^2}\int_0^{\frac{\pi}{2}} \mathrm{e}^{-\frac{2R}{\pi}\theta}\mathrm{d}\theta$$

$$= \frac{\pi R}{(R^2-1)^2}\left(1-\mathrm{e}^{-R}\right) \to 0,$$

$$\left|\int_{\gamma_r} f(z)\mathrm{d}z\right| \leqslant \frac{r}{(1-r^2)^2}\cdot \pi r = \frac{\pi r^2}{(1-r^2)^2} \to 0,$$

$$\left(\int_{[r,R]} + \int_{[-R,-r]}\right) f(z)\mathrm{d}z \to \int_{-\infty}^{+\infty} \frac{x\mathrm{e}^{\mathrm{i}x}}{(x^2+1)^2}\mathrm{d}x,$$

故

$$\begin{aligned}
\int_0^{+\infty} \frac{x\sin x}{(x^2+1)^2}\mathrm{d}x &= \frac{1}{2}\int_{-\infty}^{+\infty} \frac{x\sin x}{(x^2+1)^2}\mathrm{d}x \\
&= \frac{1}{2}\mathrm{Im}\left(\int_{-\infty}^{+\infty} \frac{x\mathrm{e}^{\mathrm{i}x}}{(x^2+1)^2}\mathrm{d}x\right) \\
&= \frac{1}{2}\mathrm{Im}\left(2\pi\mathrm{i}\mathrm{Res}(f,\mathrm{i})\right) = \pi\mathrm{Re}\left(\mathrm{Res}(f,\mathrm{i})\right) \\
&= \pi\mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{z\mathrm{e}^{\mathrm{i}z}}{(z+\mathrm{i})^2}\right)\Big|_{z=\mathrm{i}}\right) = \frac{\pi}{4\mathrm{e}}.
\end{aligned}$$

6.3. 设 $f(z)$ 和 $f^2(z)$ 在 D 内复调和, 求证: $f(z)$ 或 $\overline{f(z)}$ 在 D 内解析.

解答: 设 $f(z)=u(z)+\mathrm{i}v(z)$, 则 $f^2(z)=u^2(z)-v^2(z)+2\mathrm{i}u(z)v(z)$, 故 $\Delta u=\Delta v=\Delta(u^2-v^2)=\Delta(uv)=0$, 从而

$$\Delta u=\Delta v=0, \quad \frac{\partial u}{\partial z}\cdot\frac{\partial u}{\partial \overline{z}} = \frac{\partial v}{\partial z}\cdot\frac{\partial v}{\partial \overline{z}}, \quad \frac{\partial u}{\partial z}\cdot\frac{\partial v}{\partial \overline{z}} = -\frac{\partial v}{\partial z}\cdot\frac{\partial u}{\partial \overline{z}},$$

再注意到

$$\overline{\left(\frac{\partial u}{\partial z}\right)} = \frac{\partial u}{\partial \overline{z}}, \quad \overline{\left(\frac{\partial v}{\partial z}\right)} = \frac{\partial v}{\partial \overline{z}},$$

因此若 $\frac{\partial u}{\partial z}, \frac{\partial u}{\partial \overline{z}}, \frac{\partial v}{\partial z}, \frac{\partial v}{\partial \overline{z}}$ 中至少有一个偏导数为零, 则由 $\frac{\partial u}{\partial z}\cdot\frac{\partial u}{\partial \overline{z}} = \frac{\partial v}{\partial z}\cdot\frac{\partial v}{\partial \overline{z}}$ 可得这四个偏导数均为零; 若这四个偏导数均不为零, 将 $\frac{\partial u}{\partial z}\cdot\frac{\partial u}{\partial \overline{z}} = \frac{\partial v}{\partial z}\cdot\frac{\partial v}{\partial \overline{z}}, \frac{\partial u}{\partial z}\cdot\frac{\partial v}{\partial \overline{z}} = -\frac{\partial v}{\partial z}\cdot\frac{\partial u}{\partial \overline{z}}$ 这两个等式相乘得 $\left(\frac{\partial u}{\partial z}\right)^2 = -\left(\frac{\partial v}{\partial z}\right)^2$, 即 $\frac{\partial u}{\partial z} = \pm\mathrm{i}\frac{\partial v}{\partial z}$, 故总有 $\frac{\partial u}{\partial z} = \pm\mathrm{i}\frac{\partial v}{\partial z}$ 成立, 再由 u,v 的连续性知必有 $\frac{\partial u}{\partial z} \equiv \mathrm{i}\frac{\partial v}{\partial z}$ 或者 $\frac{\partial u}{\partial z} \equiv -\mathrm{i}\frac{\partial v}{\partial z}$, 即 $\frac{\partial f}{\partial \overline{z}} \equiv 0$ 或者 $\frac{\partial \overline{f}}{\partial \overline{z}} \equiv 0$, 也即 $f(z)$ 或 $\overline{f(z)}$ 在 D 内解析.

6.5. 设 $f(z)$ 在 D 内解析且没有零点, 求证: $\ln|f(z)|$ 在 D 内调和.

解答: 由于 $f(z)$ 在 D 内解析, 故 $\frac{\partial f(z)}{\partial \overline{z}} = \frac{\partial \overline{f(z)}}{\partial z} = 0$, 故

$$\begin{aligned}
\Delta\ln|f(z)| &= \frac{1}{2}\Delta\ln\left(f(z)\overline{f(z)}\right) = 2\frac{\partial}{\partial \overline{z}}\left(\frac{\partial}{\partial z}\ln\left(f(z)\overline{f(z)}\right)\right) \\
&= 2\frac{\partial}{\partial \overline{z}}\left(\frac{1}{f(z)\overline{f(z)}}\left(\frac{\partial f(z)}{\partial z}\overline{f(z)} + f(z)\frac{\partial \overline{f(z)}}{\partial z}\right)\right) \\
&= 2\frac{\partial}{\partial \overline{z}}\left(\frac{1}{f(z)}\frac{\partial f(z)}{\partial z}\right) \\
&= 2\left(-\frac{1}{f^2(z)}\frac{\partial f(z)}{\partial \overline{z}}\frac{\partial f(z)}{\partial z} + \frac{1}{f(z)}\frac{\partial^2 f(z)}{\partial z\partial \overline{z}}\right) = 0,
\end{aligned}$$

从而 $\ln|f(z)|$ 在 D 内调和.

6.8. 设 $u(z)$ 在 $\mathbb{C}$ 上调和且不为常数, 求证: $u(z)$ 在 $\mathbb{C}$ 上无上界也无下界.

解答: 取整函数 $f(z)$ 使得 $\mathrm{Re}\ f(z)=u(z)$. 由于 $u(z)$ 不为常数, 故 $f(z)$ 也不为常数, 从而由 **Weierstrass 定理**得 $f(\mathbb{C})$ 在 $\mathbb{C}$ 上稠密, 因此 $u(\mathbb{C})=\mathrm{Re}\ f(\mathbb{C})$ 在 $\mathbb{R}$ 上稠密, 进而 $u(z)$ 在 $\mathbb{C}$ 上无上界也无下界.

6.12. 设 $f(z)$ 是整函数, 且 $\lim\limits_{z\to\infty}\dfrac{\mathrm{Re}\ f(z)}{z}=0$, 求证: $f(z)$ 为常数.
解答: 任取定 $z\in\mathbb{C}$ 和 $R>|z|$, 由 **Cauchy 定理**得

$$0=\int_{|\omega|=R}\frac{R^2f(\omega)}{(R^2-\overline{z}\omega)^2}\mathrm{d}\omega=\int_{|\omega|=R}\frac{\omega\overline{\omega}f(\omega)}{(\omega\overline{\omega}-\overline{z}\omega)^2}\mathrm{i}\omega\mathrm{d}\theta=\mathrm{i}\int_{|\omega|=R}\frac{\overline{\omega}f(\omega)}{(\overline{z}-\overline{\omega})^2}\mathrm{d}\theta,\ (\omega=R\mathrm{e}^{\mathrm{i}\theta})$$

再取共轭得

$$0=-\mathrm{i}\int_{|\omega|=R}\frac{\omega\overline{f(\omega)}}{(z-\omega)^2}\mathrm{d}\theta=-\int_{|\omega|=R}\frac{\overline{f(\omega)}}{(z-\omega)^2}\mathrm{d}\omega,\ (\omega=R\mathrm{e}^{\mathrm{i}\theta})$$

故

$$\frac{1}{2\pi\mathrm{i}}\int_{|\omega|=R}\frac{\overline{f(\omega)}}{(z-\omega)^2}\mathrm{d}\omega=0,$$

又由于

$$\frac{1}{2\pi\mathrm{i}}\int_{|\omega|=R}\frac{f(\omega)}{(z-\omega)^2}\mathrm{d}\omega=f'(z),$$

两式相加得

$$f'(z)=\frac{1}{2\pi\mathrm{i}}\int_{|\omega|=R}\frac{f(\omega)+\overline{f(\omega)}}{(z-\omega)^2}\mathrm{d}\omega=\frac{1}{\pi\mathrm{i}}\int_{|\omega|=R}\frac{\mathrm{Re}\ f(\omega)}{(z-\omega)^2}\mathrm{d}\omega,$$

从而

$$\begin{aligned}|f'(z)|&=\frac{1}{\pi}\left|\int_{|\omega|=R}\frac{\mathrm{Re}\ f(\omega)}{\omega}\frac{\omega}{(z-\omega)^2}\mathrm{d}\omega\right|\\&\leqslant\frac{1}{\pi}\int_{|\omega|=R}\frac{|\omega|}{(|\omega|-|z|)^2}\mathrm{d}\omega\cdot\max_{|\omega|=R}\left|\frac{\mathrm{Re}\ f(\omega)}{\omega}\right|\\&=\frac{1}{\pi}\cdot\frac{R}{(R-|z|)^2}\cdot2\pi R\cdot\max_{|\omega|=R}\left|\frac{\mathrm{Re}\ f(\omega)}{\omega}\right|\\&=\frac{2R^2}{(R-|z|)^2}\cdot\max_{|\omega|=R}\left|\frac{\mathrm{Re}\ f(\omega)}{\omega}\right|\to0,\qquad\text{当 }R\to+\infty,\end{aligned}$$

因此 $f'(z)\equiv0,\forall\,z\in\mathbb{C}$, 即 $f(z)$ 为常数.